



Minibatch MCMC Algorithms for Fitting Gaussian Processes

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Abstract

Gaussian processes (GPs) are a highly flexible, nonparametric statistical model commonly used to account for correlation between observations. However, the computational load of fitting a Gaussian process is $\mathcal{O}(n^3)$ making them infeasible for large datasets. We outline both approximate and exact minibatch MCMC algorithms that substantially reduce the computation of fitting a GP by only considering small subsets of the data at a time.

Preliminaries

$Y(\mathbf{s})$, measured at locations $\mathbf{s} \in \mathcal{D} \subset \mathbb{R}^d$ is said to follow a Gaussian process (GP) if for any finite collection of locations $\mathbf{s}_1, \dots, \mathbf{s}_n$

$$\mathbf{Y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad (1)$$

where $\mathbf{Y} = (Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_n))'$ and $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the multivariate normal distribution with mean vector $\boldsymbol{\mu} = (\mu(\mathbf{s}_1), \dots, \mu(\mathbf{s}_n))'$ and covariance matrix $\boldsymbol{\Sigma} = \{\sigma_{ij}\}_{i,j=1}^n$. The mean vector is typically taken to be $\mu(\mathbf{s}) = \mathbf{x}'(\mathbf{s})\boldsymbol{\beta}$ where $\mathbf{x}(\mathbf{s}) = (x_0(\mathbf{s}), \dots, x_P(\mathbf{s}))'$ are covariates and $\boldsymbol{\beta} = (\beta_0, \dots, \beta_P)'$ are linear coefficients. The covariance is governed by a covariance function $K(\cdot)$ with parameter vetor $\boldsymbol{\phi} = (\phi_1, \dots, \phi_Q)'$ such that

$$\sigma_{ij} = K(\mathbf{s}_i, \mathbf{s}_j \mid \boldsymbol{\phi}). \quad (2)$$

If σ^2 is the total variance and $\rho(\mathbf{s}_i, \mathbf{s}_j)$ is a positive definite correlation function parameterized by $\boldsymbol{\phi}$, we write the covariance matrix as $\boldsymbol{\Sigma} = \sigma^2 \mathbf{R} = \sigma^2(\omega \mathbf{I} + (1-\omega)\mathbf{M})$ where $\mathbf{M} = \{\rho(\mathbf{s}_i, \mathbf{s}_j \mid \boldsymbol{\phi})\}_{i,j=1}^n$.

Because the joint distribution of $\mathbf{Y}$ is multivariate Gaussian, the likelihood for $\boldsymbol{\Theta} = (\boldsymbol{\beta}, \sigma^2, \omega, \boldsymbol{\phi})'$ can be written as a series of conditional distributions:

$$\mathcal{L}(\boldsymbol{\Theta}) = f_1(Y(\mathbf{s}_1) \mid \boldsymbol{\Theta}) \prod_{i=2}^n f_i(Y(\mathbf{s}_i) \mid Y(\mathbf{s}_1), \dots, Y(\mathbf{s}_{i-1}), \boldsymbol{\Theta}) \quad (3)$$

with mean μ_i and variance $\sigma^2 v_i$. Specifically, the mean and variance terms are given by:

$$\mu_i = \begin{cases} \mathbf{x}'(\mathbf{s}_1)\boldsymbol{\beta} & \text{if } i = 1 \\ \mathbf{x}'(\mathbf{s}_i)\boldsymbol{\beta} + \mathbf{R}(i, \mathcal{P}_i)\mathbf{R}^{-1}(\mathcal{P}_i, \mathcal{P}_i)(\mathbf{Y}_{\mathcal{P}_i} - \mathbf{X}_{\mathcal{P}_i}\boldsymbol{\beta}) & \text{if } i > 1 \end{cases} \quad (4)$$

$$v_i = \begin{cases} 1 & \text{if } i = 1 \\ 1 - \mathbf{R}(i, \mathcal{P}_i)\mathbf{R}^{-1}(\mathcal{P}_i, \mathcal{P}_i)\mathbf{R}(\mathcal{P}_i, i) & \text{if } i > 1 \end{cases} \quad (5)$$

where $\mathcal{P}_i = \{1, \dots, i-1\}$ is the set of points preceding observation i .

μ_i and v_i in (4) and (5) still require dealing with large matrices through $\mathbf{R}(\mathcal{P}_i, \mathcal{P}_i)$ because $\#\mathcal{P}_i$ grows as $i \rightarrow n$. Hence, we adopt the Vecchia process approximation framework by replacing $\mathcal{P}_i$ in (4) and (5) with the set $\mathcal{N}_i$ of the M nearest neighbors of $\mathbf{s}_i$.

Conjugate Parameters

In the Bayesian paradigm, we assume priors such that the conditional distribution of each β_p and σ^2 are conjugate with respect to the Vecchia likelihood in (3) and are given by

$$\beta_p \mid - \sim \mathcal{N}\left(\left[\frac{\sum_{i=1}^n q_1(\mathbf{s}_i)}{\sigma^2} + \frac{1}{s_p^2}\right]^{-1} \left(\frac{\sum_{i=1}^n q_2(\mathbf{s}_i)}{\sigma^2} + \frac{m_p}{s_p^2}\right), \left[\frac{\sum_{i=1}^n q_1(\mathbf{s}_i)}{\sigma^2} + \frac{1}{s_p^2}\right]^{-1}\right) \quad (6)$$

$$\sigma^2 \mid - \sim \mathcal{IG}\left(\frac{n}{2} + a_\sigma, \frac{\sum_{i=1}^n q_3(\mathbf{s}_i)}{2} + b_\sigma\right) \quad (7)$$

with “ $-$ ” denoting all other parameters and data, and $q_1(\mathbf{s}_i)$, $q_2(\mathbf{s}_i)$, and $q_3(\mathbf{s}_i)$ are

$$q_1(\mathbf{s}_i) = \frac{1}{v_i} \left[x_p(\mathbf{s}_i) - \mathbf{R}(i, \mathcal{N}_i)\mathbf{R}^{-1}(\mathcal{N}_i, \mathcal{N}_i)\mathbf{X}_{\mathcal{N}_i, p} \right]^2 \quad (8)$$

$$q_2(\mathbf{s}_i) = \frac{1}{v_i} \left[x_p(\mathbf{s}_i) - \mathbf{R}(i, \mathcal{N}_i)\mathbf{R}^{-1}(\mathcal{N}_i, \mathcal{N}_i)\mathbf{X}_{\mathcal{N}_i, p} \right] r_p(\mathbf{s}_i) \quad (9)$$

$$q_3(\mathbf{s}_i) = \frac{1}{v_i} [Y(\mathbf{s}_i) - \mu_i]^2 \quad (10)$$

where $\mathbf{X}_{\mathcal{N}_i, p}$ is the p^{th} column of $\mathbf{X}_{\mathcal{N}_i}$ and $\mathbf{X}_{\mathcal{N}_i, -p}$ is all columns *except* the p^{th} column and

$$r_p(\mathbf{s}_i) = Y(\mathbf{s}_i) - \mathbf{x}'_{-p}(\mathbf{s}_i)\boldsymbol{\beta}_{-p} - \mathbf{R}(i, \mathcal{N}_i)\mathbf{R}^{-1}(\mathcal{N}_i, \mathcal{N}_i)(\mathbf{Y}_{\mathcal{N}_i} - \mathbf{X}_{\mathcal{N}_i, -p}\boldsymbol{\beta}_{-p}).$$

Non-conjugate Parameters

We generally assume $\boldsymbol{\theta} = (\omega, \boldsymbol{\phi})' \sim \pi(\cdot)$. Due to lack of conjugacy, MCMC simulation of $\boldsymbol{\theta}$ is done via accept-reject algorithms with acceptance rule for a proposal $\boldsymbol{\theta}_{\text{prop}}$ given $\boldsymbol{\theta}_{\text{cur}}$ as

$$\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}_{\text{cur}}) = \sum_{i=1}^n \Lambda_i - \log \left(\frac{\pi(\boldsymbol{\theta}_{\text{prop}})g(\boldsymbol{\theta}_{\text{cur}} \mid \boldsymbol{\theta}_{\text{prop}})}{\pi(\boldsymbol{\theta}_{\text{cur}})g(\boldsymbol{\theta}_{\text{prop}} \mid \boldsymbol{\theta}_{\text{cur}})} \right) + L \quad (11)$$

where $\Lambda_i = \log[f_i(Y(\mathbf{s}_i) \mid \mathbf{Y}_{\mathcal{N}_i}, \boldsymbol{\beta}, \sigma^2, \boldsymbol{\theta}_{\text{prop}})/f_i(Y(\mathbf{s}_i) \mid \mathbf{Y}_{\mathcal{N}_i}, \boldsymbol{\beta}, \sigma^2, \boldsymbol{\theta}_{\text{cur}})]$ is the log-likelihood ratio for observation i , $g(\cdot)$ is the proposal distribution and L is a random variable. Accepting a proposed $\boldsymbol{\theta}_{\text{prop}}$ occurs if and only if $\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}_{\text{cur}}) > 0$.

Minibatch Approximations

Conjugate Parameters

$$\begin{aligned} \sum_{i=1}^n q_j(\mathbf{s}_i) &= n\bar{q}_{j,n} = n\frac{1}{n} \sum_{i=1}^n q_j(\mathbf{s}_i) \\ &\approx n\bar{q}_{j,B} = n\frac{1}{B} \sum_{i \in \mathcal{B}} q_j(\mathbf{s}_i) \end{aligned} \quad (12)$$

for $j \in \{1, 2, 3\}$ where $\mathcal{B} \subseteq \{1, \dots, n\}$ is a randomly sampled minibatch with $\#\mathcal{B} = B$. Because our approximation is an average, we assume

$$n\bar{q}_{j,B} \sim \mathcal{N}\left(n\bar{q}_{j,n}, \frac{n^2}{B} \sqrt{\frac{n-B}{n-1}} \sigma_{q_j}^2\right) \quad (13)$$

where $\sigma_{q_j}^2 = \text{Var}(q_j(\mathbf{s}_i))$.

Non-conjugate Parameters

$$\begin{aligned} \sum_{i=1}^n \Lambda_i &= n\bar{\Lambda}_n = n\frac{1}{n} \sum_{i=1}^n \Lambda_i \\ &\approx n\bar{\Lambda}_B = n\frac{1}{B} \sum_{i \in \mathcal{B}} \Lambda_i \end{aligned} \quad (14)$$

where $\mathcal{B} \subseteq \{1, \dots, n\}$ is a random minibatch of the observations and $\#\mathcal{B} = B$. Because our minibatch approximation is an average, we assume

$$n\bar{\Lambda}_B \sim \mathcal{N}\left(n\bar{\Lambda}_n, \frac{n^2}{B} \sqrt{\frac{n-B}{n-1}} \sigma_{\Lambda}^2\right) \quad (15)$$

where $\sigma_{\Lambda}^2 = \text{Var}(\Lambda_i)$.

Algorithms 1 and 2 below outline two possible minibatch MCMC algorithms for fitting GPs to data. Algorithm 1 uses a minibatch approximation of the Metropolis-Hastings rule, by approximating the Metropolis acceptance probability using a fixed proportion of the data. Algorithm 2 uses the Barker accept-reject rule, giving draws from the posterior distribution using the full dataset, where the required minibatch size (B) will depend on the proposal $\boldsymbol{\theta}_{\text{prop}}$.

Algorithm 1

- 1: Initialize $\boldsymbol{\beta}_{\text{cur}}, \sigma_{\text{cur}}^2, \boldsymbol{\theta}_{\text{cur}}$
- 2: Set batch size B
- 3: Randomly divide the n datapoints into H batches of size B
- 4: **for** epoch in $1, \dots, E$ **do**
- 5: **for** h in $1, \dots, H$ **do**
- 6: **for** p in $0, \dots, P$ **do**
- 7: Draw β_p from (6) using minibatch h in (12)
- 8: **end for**
- 9: Draw σ^2 from (7) using minibatch h in (12)
- 10: Propose $\boldsymbol{\theta}_{\text{prop}} \sim g(\cdot)$
- 11: Let $L = -\log(U)$, $U \sim \mathcal{U}(0, 1)$. Using minibatch h calculate

$$\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}_{\text{cur}}) = n\bar{\Lambda}_B + \log \left(\frac{\pi(\boldsymbol{\theta}_{\text{prop}})g(\boldsymbol{\theta} \mid \boldsymbol{\theta}_{\text{prop}})}{\pi(\boldsymbol{\theta}_{\text{cur}})g(\boldsymbol{\theta}_{\text{prop}} \mid \boldsymbol{\theta})} \right) + L$$

- 12: **$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_{\text{prop}}$** if $\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}) > 0$ using the approximated acceptance rule
- 13: Store $\boldsymbol{\beta}, \sigma^2, \boldsymbol{\theta}$

Algorithm 2

- 1: Initialize $\boldsymbol{\beta}_{\text{cur}}, \sigma_{\text{cur}}^2, \boldsymbol{\theta}_{\text{cur}}$
- 2: Choose B_β and B_{σ^2} for minibatch approximations
- 3: Set initial batch size B_{init} and increments B_{inc} for Barker test
- 4: Set logistic approximation cutoff $c \leq 3$
- 5: Estimate correction distribution $h(x \mid c)$ via penalized least squares
- 6: **for** i in $1, \dots, I$ **do**
- 7: **for** p in $0, \dots, P$ **do**
- 8: Sample minibatch $\mathcal{B}_\beta$ of size B_β from $\{1, \dots, n\}$
- 9: Draw β_p from (6) using the minibatch approximation in (12)
- 10: **end for**
- 11: Sample minibatch $\mathcal{B}_{\sigma^2}$ of size B_{σ^2} from $\{1, \dots, n\}$
- 12: Draw σ^2 from (7) using the minibatch approximation in (12)
- 13: Propose $\boldsymbol{\theta}_{\text{prop}} \sim g(\cdot)$
- 14: Sample $\mathcal{B}_\theta$ of size B_{init} from $\{1, \dots, n\}$
- 15: Set $\sigma_\Lambda^2 = \text{Var}(\{\Lambda_i\}_{i \in \mathcal{B}_\theta})$
- 16: **while** $(n^2/B_\theta)\sqrt{(n-B_\theta)/(n-1)}\sigma_\Lambda^2 > c$ **do**
- 17: Append $\mathcal{B}_\theta$ by sampling an additional B_{inc} from $\{1, \dots, n\} \setminus \mathcal{B}_\theta$
- 18: Set $B_\theta \leftarrow B_\theta + B_{\text{inc}}$
- 19: Recalculate $\sigma_\Lambda^2 = \text{Var}(\{\Lambda_i\}_{i \in \mathcal{B}_\theta})$
- 20: **end while**
- 21: Draw $L_2 \sim h(\cdot \mid c)$
- 22: Draw $L_1^* \sim \mathcal{N}(0, c - (n^2/B_\theta)\sqrt{(n-B_\theta)/(n-1)}\sigma_\Lambda^2)$
- 23: Calculate

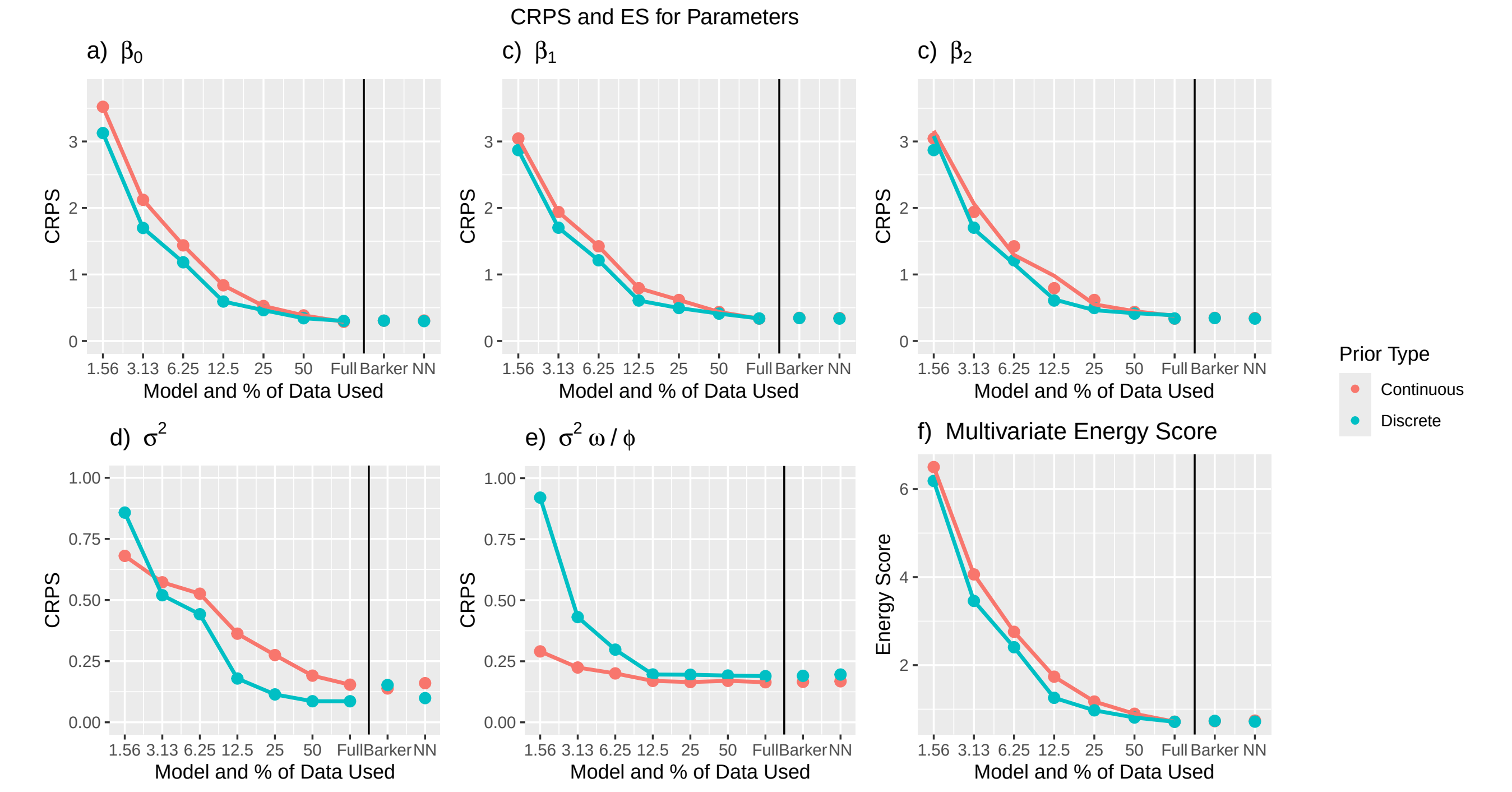
$$\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}) = n\bar{\Lambda}_B + \log \left(\frac{\pi(\boldsymbol{\theta}_{\text{prop}})g(\boldsymbol{\theta} \mid \boldsymbol{\theta}_{\text{prop}})}{\pi(\boldsymbol{\theta})g(\boldsymbol{\theta}_{\text{prop}} \mid \boldsymbol{\theta})} \right) + L_1^* + L_2$$

- 24: **$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_{\text{prop}}$** if $\Delta(\boldsymbol{\theta}_{\text{prop}}, \boldsymbol{\theta}) > 0$ using the approximated acceptance rule
- 25: Store $\boldsymbol{\beta}, \sigma^2, \boldsymbol{\theta}$

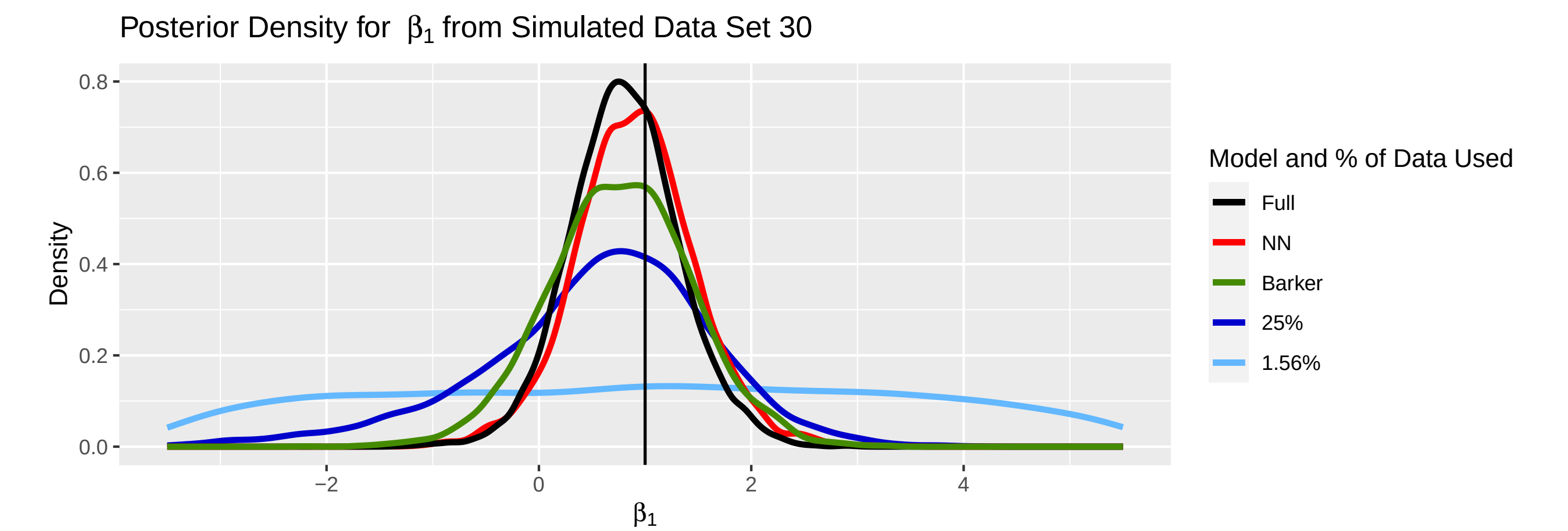
Simulation Results

We carried out a simulation study using 50 simulated data sets with $n = 8000$ and $n = 120,000$ at locations $\mathbf{s}_i$ simulated uniformly on the unit square using a stationary exponential correlation function with range ϕ and nugget ω .

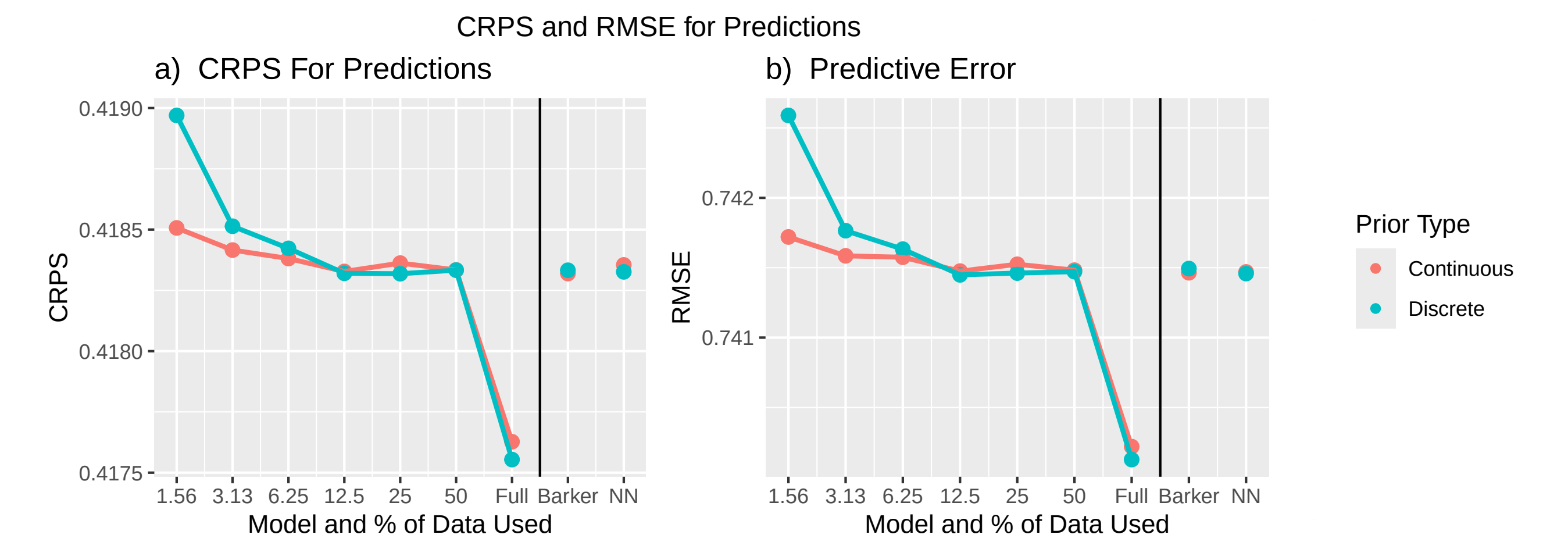
Parameter Inference



The figure below displays density plots for the β_1 parameter across various minibatch sizes. The posterior distribution for each algorithm is centered at the true value, but the variance of the posterior distribution increases as the minibatch size decreases.



Predictive Accuracy



Computational Efficiency

